

Effective Dataset Distillation for Spatio-Temporal Forecasting with Bi-dimensional Compression (Supplementary Material)

A. Training Details

In this section, we elaborate on the training details that are excluded in the main paper due to the lack of space. We set the number of columns R in the location embeddings to 10 for all methods. When using the location encoder model, the hidden dimension is set to 32. We set the batch size to 256 for GBA [1], 128 for GLA [1], 128 for ERA5 [2], 64 for CA [1], and 64 for CAMS [3] datasets. When evaluating the distilled dataset, we train MTGNN for 400 epochs for the GBA and GLA datasets, and 100 epochs for the other datasets. This also holds for evaluating the datasets distilled by the baselines.

B. Hyperparameter Settings for Baselines

The hyperparameter settings of the baseline experiments that were not mentioned in our paper are mentioned in this section. We don't split the locations into subsets (K) when training the MTGNN [4] model on the synthetic dataset distilled by baselines, since this is the default option in the paper and the authors' code. When distilling with DC [5], we set the batch size to 128 for GBA, 64 for GLA, 64 for ERA5, 32 for CA, and 32 for CAMS datasets. Note that we use the maximum batch size that could be used while increasing the batch size by a factor of 2.

For MTT [6], we use the MTGNN model to generate training trajectories. A total of 20 expert buffers are created, and the trajectory in each buffer is generated by training for 15 epochs. While distilling with MTT, we set the "maximum start epoch" to 4, "number of expert epoch" to 2, and "number of synthetic steps" to 10, which was the most empirically effective combination in our experiments.

For CondTSF [7], DLinear [8] model was used to generate training trajectories. Similarly, 20 expert buffers were created, each trained for 15 epochs. The same hyperparameter configuration as in MTT was adopted (maximum start epoch = 4, number of expert epoch = 2, number of synthetic steps = 10) for the trajectory matching process. We set the additive update ratio to 0.01 and optimize the value term for every 3 epochs as the default setting in the paper.

For TimeDC [9], we used the model proposed in the TimeDC paper to generate training trajectories, with 20 expert buffers each trained for 40 epochs. All the other hyperparameters and settings for TimeDC experiments followed those specified in the paper.

The learning rates used to train the model on the synthetic data, to distill the synthetic data, and to train the model on the

original data for creating expert trajectories are all selected via grid search over $\{10^{-2}, 10^{-3}, 10^{-4}\}$, except for TimeDC. For TimeDC experiments, we search within a larger range, from 10^{-2} to 10^{-7} .

REFERENCES

- [1] Y. Fang, Y. Liang, B. Hui, Z. Shao, L. Deng, X. Liu, X. Jiang, and K. Zheng, "Efficient large-scale traffic forecasting with transformers: A spatial data management perspective," in *KDD*, 2025.
- [2] H. Hersbach, B. Bell, P. Berrisford, G. Biavati, A. Horányi, J. Muñoz Sabater, J. Nicolas, C. Peubey, R. Radu, I. Rozum, D. Schepers, A. Simmons, C. Soci, D. Dee, and J.-N. Thépaut, "Era5 hourly data on single levels from 1940 to present," Copernicus Climate Change Service (C3S) Climate Data Store (CDS), 2023.
- [3] A. Inness, M. Ades, A. Agustí-Panareda, J. Barré, A. Benedictow, A.-M. Blechschmidt, J. J. Dominguez, R. Engelen, H. Eskes, J. Flemming *et al.*, "The cams reanalysis of atmospheric composition," *Atmospheric Chemistry and Physics*, vol. 19, no. 6, 2019.
- [4] Z. Wu, S. Pan, G. Long, J. Jiang, X. Chang, and C. Zhang, "Connecting the dots: Multivariate time series forecasting with graph neural networks," in *KDD*, 2020.
- [5] B. Zhao, K. R. Mopuri, and H. Bilen, "Dataset condensation with gradient matching," in *ICLR*, 2021.
- [6] G. Cazenavette, T. Wang, A. Torralba, A. A. Efros, and J.-Y. Zhu, "Dataset distillation by matching training trajectories," in *CVPR*, 2022.
- [7] J. Ding, Z. Liu, G. Zheng, H. Jin, and L. Kong, "Condstf: one-line plugin of dataset condensation for time series forecasting," *NeurIPS*, 2024.
- [8] A. Zeng, M. Chen, L. Zhang, and Q. Xu, "Are transformers effective for time series forecasting?" in *AAAI*, 2023.
- [9] H. Miao, Z. Liu, Y. Zhao, C. Guo, B. Yang, K. Zheng, and C. S. Jensen, "Less is more: Efficient time series dataset condensation via two-fold modal matching," *PVLDB*, vol. 18, no. 2, pp. 226–238, 2024.